

Motivation/Main results 4 Minutes

Preliminaries 5 Minutes

Neighbourhood frame

Definition

Let X be a non-empty set and $k \geq 1$. A k -neighborhood frame (k -n-frame) is a pair $(X, \vec{\tau})$, where $\vec{\tau} = (\tau_1, \dots, \tau_k)$ and each $\tau_i : X \rightarrow 2^{2^X}$ assigns to every $x \in X$ a *filter* $\tau_i(x)$ on X , i.e.

- $\tau(x) \neq \emptyset$
- if $U \in \tau(x)$ and $U \subseteq V$ then $V \in \tau(x)$
- if $U, V \in \tau(x)$ then $U \cup V \in \tau(x)$

Full Product of Frames

Definition

Let $\mathcal{X} = (X, \tau_1)$ and $\mathcal{Y} = (Y, \tau_2)$ be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame $\mathcal{X} \times_n \mathcal{Y} = (\mathbf{X} \times \mathbf{Y}, \tau'_1, \tau'_2, \tau)$ defined as follows :

$$\tau'_1(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\},$$

$$\tau'_2(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\},$$

$$\tau(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) \exists W \in \tau_2(y) : V \times W \subseteq U\}$$

Bounded morphism

Definition

Let $F = (W, \tau_1, \dots, \tau_k)$ and $F' = (W', \sigma_1, \dots, \sigma_k)$ be neighborhood frames. A function $f : W \rightarrow W'$ is called a bounded morphism if

- for all $1 \leq i \leq k$, for all $w \in W$ and all $U \in \tau_i(w)$, there is a $V \in \sigma_i(f(w))$ with $V = f(U)$,
- for all $1 \leq i \leq k$, for all $w \in W$ and all $V \in \sigma_i(f(w))$, there exists $U \in \tau_i(w)$ such that $f(U) \subseteq V$.

Note: If $F \twoheadrightarrow F'$, then $\text{Log}(\{F\}) \subseteq \text{Log}(\{F'\})$.

Definition

Let L_1 and L_2 be two unimodal logics. Then the *fusion* ($\otimes$) of these logics is

$$K_2 + (L_1)_{\Box \rightarrow \Box_1} + (L_2)_{\Box \rightarrow \Box_2}$$

Furthermore the full product ($\times_n^+$) is defined as

$$L_1 \times_n^+ L_2 = \text{Log}\{\mathcal{X} \times \mathcal{Y} \mid \mathcal{X} \models L_1 \text{ and } \mathcal{Y} \models L_2\}$$

Key ideas sketched 15 Minutes

Main Theorem

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \otimes \Lambda \otimes \Lambda + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p = \Lambda \times_n^+ \Lambda$$

In the following we will sketch the proof for $\Lambda = T$.

We denote $T \otimes T \otimes T + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$ as T_3m

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3m$

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $Log(T_{\omega,\omega,\omega}^{r,m}) = T_3m$
- Note: For any Kripke frame $F = (W, R_1, \dots, R_n)$, we can construct a neighborhood frame F' , i.e. for any $w \in W$: $\tau(w) = \{U \mid R_1(w) \subseteq U \subseteq W\}$ where $Log(F) = Log(F')$

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $Log(T_{\omega,\omega,\omega}^{r,m}) = T_3m$
- Note: For any Kripke frame $F = (W, R_1, \dots, R_n)$, we can construct a neighborhood frame F' , i.e. for any $w \in W$: $\tau(w) = \{U \mid R_1(w) \subseteq U \subseteq W\}$ where $Log(F) = Log(F')$
- construct neighbourhood product $N_{\omega}^r \times_n^+ N_{\omega}^r$ s.t. there is surjective bounded morphism from $N_{\omega}^r \times_n^+ N_{\omega}^r$ to $N(T_{\omega,\omega,\omega}^{r,m})$

$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

$$\text{Log}(T_{\omega, \omega, \omega}^{r, m}) = T_3 m$$

- Pick $\mathbb{C} = \{F \mid F \models T_3 m\}$

Lemma

$$\text{Log}(\mathbb{C}) = T_3 m$$

- By Sahlqvist Theorem

$$\text{Log}(T_{\omega, \omega, \omega}^{r, m}) = T_3 m$$

- Pick $\mathbb{C} = \{F \mid F \models T_3 m\}$

Lemma

$$\text{Log}(\mathbb{C}) = T_3 m$$

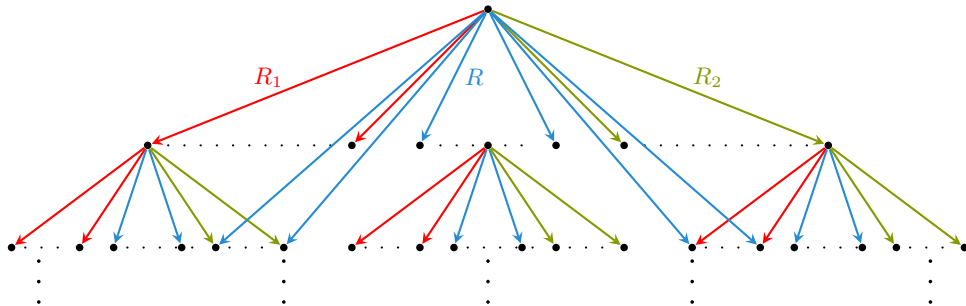
- By Sahlqvist Theorem

Lemma

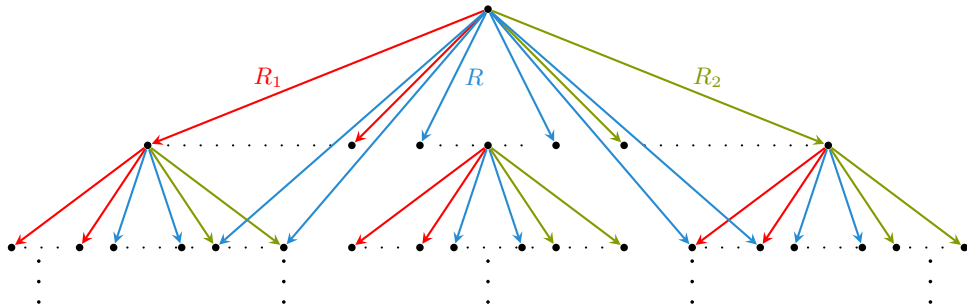
$T_3 m$ has the finite model property

- By Filtration Theorem

$$\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3 m$$



$$\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3m$$



- Note: $\Box p \rightarrow \Box_1 p \wedge \Box_2 p$ is derivable
- $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) \subseteq T_3m$ shown by unraveling

Definition of the neighborhood frame N_w^r

Definition

Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0a_1a_2... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences.

Definition of the neighborhood frame N_w^r

Definition

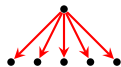
Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0a_1a_2... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences. Let $\alpha = a_0a_1a_2... \in X$. Then

$$U_k(\alpha) = \{\beta \in X \mid \text{forg}(\alpha)R\text{forg}(\beta) \text{ and } \alpha \upharpoonright_m = \beta \upharpoonright_m, \text{ with } m = \max(k, st(\alpha))\}$$

where $\text{forg}(\alpha)$ deletes all $\star$ and $st(\alpha) = \min\{N \mid \forall k \geq N : a_k = \star\}$



Open questions 1 Minute
